

PART I.

1. A program that uses bitwise operators.

```
1 package mypackage;
2
3 public class Part1 {
4
5     public static void main(String[] args) {
6         int a = 15, b = 7;
7
8         System.out.println(a++ | ++b);
9     }
10 }
```

Output:

```
15
```

2. A program that uses the ternary operator (? :).

```
1 package mypackage;
2
3 public class Part2 {
4
5     public static void main(String[] args) {
6         int a = 10, b = 20;
7
8         int max = (a > b) ? a : b;
9
10        System.out.println("Largest number is " + max);
11    }
12 }
13 }
```

Output:

```
Largest number is 20
```

3. A program that uses assignment operators (+=, -=, *=, /=).

```
1 package mypackage;
2
3 public class Part3 {
4
5     public static void main(String[] args) {
6         int num = 20;
7
8         System.out.println("Initial value: " + num);
9
10        // += operator
11        num += 10;
12        System.out.println("After += 10: " + num);
13
14        // -= operator
15        num -= 5;
16        System.out.println("After -= 5: " + num);
17
18        // *= operator
19        num *= 2;
20        System.out.println("After *= 2: " + num);
21
22        // /= operator
23        num /= 5;
24        System.out.println("After /= 5: " + num);
25    }
26 }
27
```

Output:

```
Initial value: 20
After += 10: 30
After -= 5: 25
After *= 2: 50
After /= 5: 10
```

Part II.

Create one Java program that demonstrates the use of multiple operators.

```
1 package mypackage;
2
3 public class PartII {
4
5     public static void main(String[] args) {
6         int p = 12;
7         int q = 5;
8         int r = 8;
9         int s = 3;
10        int t = 6;
11
12        // 1. Arithmetic operator
13        System.out.println("1. Arithmetic result: " + (p + q * 2));
14
15        // 2. Assignment operator
16        r += 4;
17        System.out.println("2. Assignment result: " + r);
18
19        // 3. Bitwise operator
20        System.out.println("3. Bitwise result: " + (t & s));
21
22        // 4. Relational operator (boolean)
23        System.out.println("4. Relational result: " + (p > q));
24
25        // 5. Logical operator (boolean)
26        System.out.println("5. Logical result: " + ((p > q) && (r >= 10)));
27
28        // 6. Unary tricky increment expression
29        int x = 7;
30        System.out.println("6. Tricky unary result: " + (x++ + ++x));
31    }
32 }
33
```

Output:

```
1. Arithmetic result: 22
2. Assignment result: 12
3. Bitwise result: 2
4. Relational result: true
5. Logical result: true
6. Tricky unary result: 16
```